

AI Workflow Design

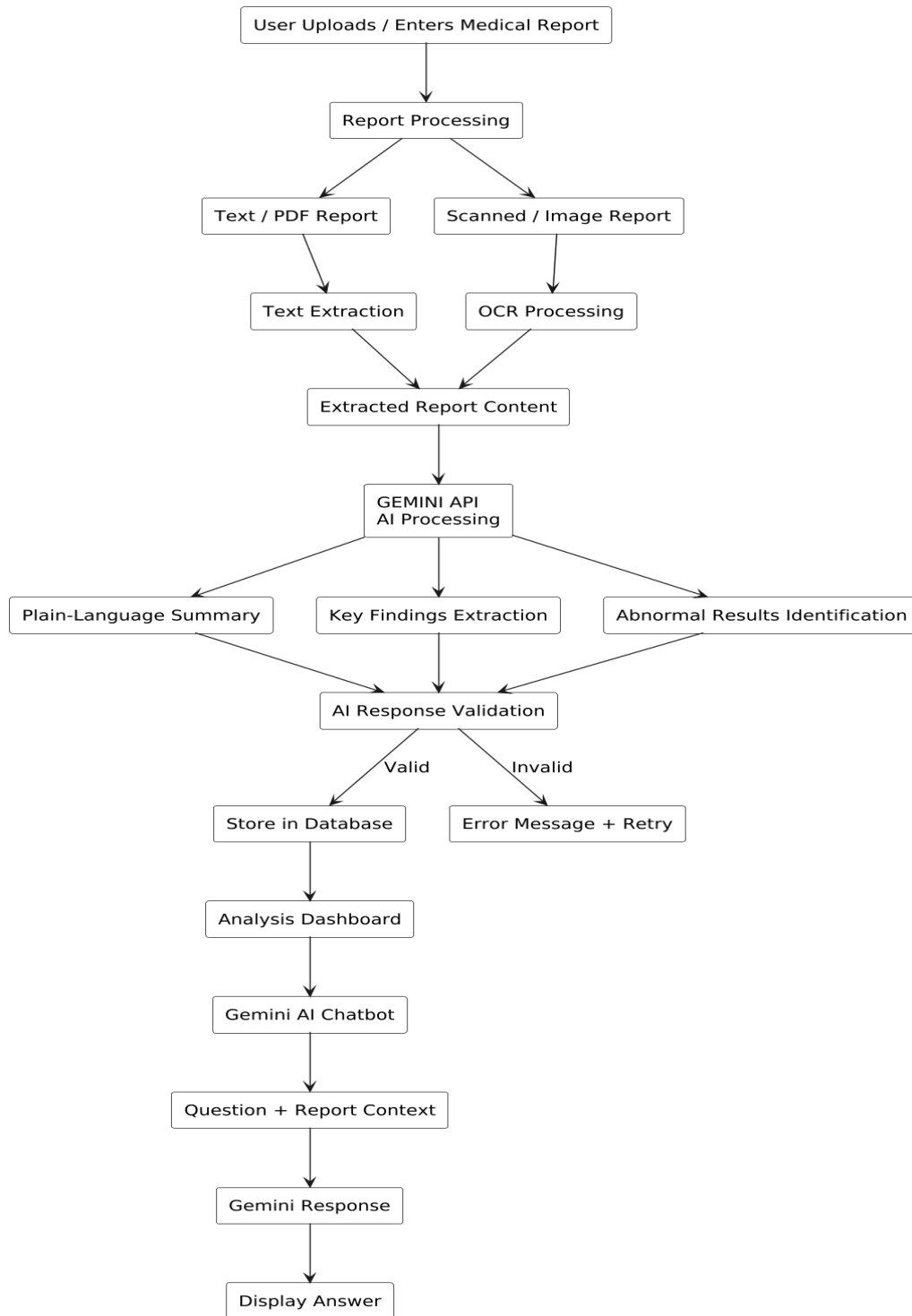


Fig: AI Workflow Design

AI Workflow Design Briefly Described

1. User Uploads / Enters Medical Report

The user uploads a medical report or enters the report information into the system.

2. Report Processing

The system processes the submitted report and determines how its content should be extracted.

3. Text / PDF Report

For digital or text-based reports, the existing text content is processed directly.

4. Scanned / Image Report

For scanned or image-based reports, OCR processing is used to extract readable text from the document.

5. Text Extraction

The system extracts machine-readable text from the submitted report.

6. OCR Processing

OCR converts text contained in scanned documents or images into machine-readable report content.

7. Extracted Report Content

The extracted content is prepared and passed to the AI system for further analysis.

8. Gemini API — AI Processing

The extracted report content is sent to the **Google Gemini API**, where Gemini analyzes the medical information according to the application's prompts.

9. Plain-Language Summary

Gemini generates a simplified summary of the medical report so that users can understand the information more easily.

10. Key Findings Extraction

Gemini identifies the important medical and laboratory findings contained in the report.

11. Abnormal Results Identification

Gemini identifies potentially abnormal values or findings that may require the user's attention.

12. AI Response Validation

The application validates the Gemini response to ensure that the generated output is complete, properly structured, and usable.

13. Valid Response — Store in Database

If the response is valid, the generated analysis is stored in the database for future access.

14. Invalid Response — Error Message + Retry

If the AI response is invalid or the AI service encounters an error, the system displays an appropriate error message and allows the process to be retried.

15. Analysis Dashboard

The validated AI-generated analysis is displayed on the user's medical report analysis dashboard.

16. Gemini AI Chatbot

The user can interact with the Gemini-powered chatbot to ask questions about their medical report.

17. Question + Report Context

The user's question is combined with the relevant report context so that Gemini can provide a context-aware answer.

18. Gemini Response

Gemini processes the question and report context and generates an appropriate response.

19. Display Answer

The generated response is displayed to the user through the AI chatbot interface.